Example 4.22 What is the unit digit of the sum $2! + 3! + 4! + ... + 22!$?

Solution:
From $5!$ onwards for all $n!$ the unit digit is zero and hence the contribution to the unit digit is through

$2! + 3! + 4!$ only. which is $2 + 6 + 24 = 32$. Therefore the required unit digit is 2.

Example 4.23 If $\frac{1}{7!} + \frac{1}{8!} = \frac{A}{9!}$ then find the value of A.

Solution:
We have, $\frac{A}{9 \times 8 \times 7!} = \frac{1}{7!} + \frac{1}{8 \times 7!}$

Therefore, $\frac{1}{7!} \times \frac{A}{9 \times 8} = \frac{1}{7!} \times \left[1 + \frac{1}{8}\right]$ equivalently, $\frac{A}{72} = \frac{9}{8}$, which imply $A = 81$.

Example 4.24 Prove that $\frac{(2n)!}{n!} = 2^n(1.3.5 \cdots (2n-1))$.

Solution:

$$\frac{(2n)!}{n!} = \frac{1.2.3.4 \cdots (2n-2).(2n-1)2n}{n!}$$

$$= \frac{(1.3.5. \cdots (2n-1))\,(2.4.6. \cdots (2n-2).2n)}{n!} \text{ (Grouping the odd and even numbers separately)}$$

$$= \frac{(1.3.5. \cdots (2n-1))\,(2^n)\,(1.2.3. \cdots (n-1).n)}{n!} \text{ (taking out the 2' s)}$$

$$= \frac{(1.3.5. \cdots (2n-1))\,(2^n)\,n!}{n!}$$

$$= 2^n\,(1.3.5. \cdots (2n-1))\,.$$



Exercise - 4.1

1. (i) A person went to a restaurant for dinner. In the menu card, the person saw 10 Indian and 7 Chinese food items. In how many ways the person can select either an Indian or a Chinese food?

   (ii) There are 3 types of toy car and 2 types of toy train available in a shop. Find the number of ways a baby can select a toy car and a toy train?

   (iii) How many two-digit numbers can be formed using 1,2,3,4,5 without repetition of digits?